

IMO NCSR 12

Meeting summary

3 June 2025



The 12th meeting of the IMO's Sub-Committee on Navigation, Communication and Search and Rescue (NCSR 12) was held 13-22 May 2025 at the IMO Headquarters in London. The meeting was supplemented by hybrid (online) arrangements.

LISCR participated in the following groups, in addition to the plenary:

Working Group	Subject
Working Group 1	ITU matters, Software maintenance and S-100 ECDIS
Working Group 2	WWRNS, VDES and 12 ENP
Working Group 3	GMDSS, SAR and EPIRB
Experts Group	Routeing
Drafting Group	Bei-Dou satellite system

Routeing measures

Area to be Avoided off the Coast of Réunion

NCSR 12 approved the establishment of an Area to be Avoided (ATBA) surrounding the island of Réunion, which will enter into force six months after adoption by the 111th session of the Maritime Safety Committee (MSC 111) (scheduled for May 2026).

Mandatory Ship Reporting System in the Adriatic Sea (ADRIREP)

NCSR 12 approved revisions to the mandatory ship reporting system in the Adriatic Sea. The amendments include automated electronic reporting and expanded coverage to additional States, which will enter into force six months after adoption by MSC 111 (scheduled for May 2026).

Mandatory Ship Reporting Systems in and around European Coastal Waters

A broader proposal to amend ten mandatory reporting systems in European waters, introducing a requirement to report liability insurance details, was partially agreed, i.e., only the amendments to the SOUNDREP and BELTREP systems (covering the Danish Straits) were agreed for adoption by MSC 111.

South Åland Sea Deep-Water Route

NCSR 12 approved SN.1/Circ.272/Add.3, which updates water depth information and changes the geodetic datum used on the chart for the South Åland Sea deep-water route, subject to endorsement by MSC 111.

Voluntary Traffic Separation Schemes off Peru

Peru informed NCSR 12 of its intention to establish three voluntary Traffic Separation Schemes (TSSs) off its coast. The NCSR 12 invited Peru to consider submitting a proposal to a future session for consideration of the intended measures, should they require adoption by IMO.

Long Range Identification and Tracking (LRIT)

Use of date and time parameters

NCSR 12 endorsed the LRIT Operational governance body (OGB)'s recommendations, highlighting that:

- Data Centres (DC) should monitor ship data and report equipment malfunctions to flag Administrations;
- Users identifying Global Navigation Satellite System (GNSS) errors should notify the relevant Administration via designated LRIT contact points.

Performance review and audit

NCSR 12 reviewed:

- DC and International Data Exchange (IDE) audit report, and noted that all pending audits were completed;
- System performance from Feb 2024 to Jan 2025, which was generally compliant;

Contracting Governments were urged to:

- Keep the Global Integrated Shipping Information System (GISIS) Data Distribution Plan (DDP) entries up to date;
- Address GPS rollover risks;
- Ensure DCs cooperate with the LRIT Coordinator and settle audit-related financial obligations.

NCSR 12 also noted comments on audit findings, the value of LRIT for SAR and maritime security (especially in the Gulf of Aden), and the importance of training and capacity-building.

Global Maritime Distress and Safety System (GMDSS), including dissemination of Maritime Safety Information (MSI)

NAVTEX

NCSR 12 reminded Member States to follow procedures given in the NAVTEX Manual (MSC.1/Circ.1403/Rev.2) regarding the termination of NAVTEX stations.

NAVDAT

NCSR 12 took the following decisions:

- The joint NAVTEX–NAVDAT Coordinating Panel was established as the IMO Terrestrial Broadcast Services Coordinating Panel, with revised terms of reference, and a draft MSC circular was finalised for approval;
- With regard to the discrepancy in message priority categories between NAVTEX and NAVDAT, NCSR 12 supported the mapping of priorities and proposed amendments to the NAVDAT Manual, with further review by IHO and WMO;

NCSR 12 agreed that:

- the priorities should follow the proposed mapping framework; and
- the reception of meteorological forecasts broadcast via NAVDAT should not be rejected;

Dissemination of MSI and SAR related information

NCSR 12 prepared draft amendments to SOLAS chapters IV and V to clarify that coastal States are obligated to disseminate Maritime Safety Information (MSI) through all recognized Mobile Satellite Services (RMSS), not limited to Inmarsat. The amendment reflects the expanded recognition of multiple RMSS providers under the GMDSS.

The proposed amendments were sent to MSC 110 for adoption by MSC 111, with an expected entry into force of 1 January 2028. This means that the original instruction by MSC 108, which had set a compliance deadline of 31 December 2026, was not met.

NCSR 12 also finalized a draft revision of resolution MSC.509(105) (Rev.1) on *Provision of Maritime Safety Information (MSI) and Search and Rescue (SAR) Related Information over Recognized Mobile Satellite Services (RMSSs)* to reflect these SOLAS amendments and to ensure alignment with the evolving GMDSS framework. Consequential amendments to related MSI instruments are under consideration by the International Hydrographic Organization (IHO) and the World Meteorological Organization (WMO), and the IMO/ITU Joint Experts Group was instructed to consider corresponding updates to MSC.1/Circ.1659 on *Guidance for the Dissemination of Search and Rescue Related Information through the International Enhanced Group Call Service*.

ITU matters

Draft IMO position on relevant WRC-27 agenda items

NCSR 12 addressed the development of the IMO's technical position for WRC-27 and tasked the IMO/ITU Joint Working Group to further develop the position and report back to NCSR 13. Key issues include:

- IMO emphasizes the need to protect frequencies used for maritime safety services, — including maritime mobile-satellite, VHF maritime mobile band, and radars operating in the 13.75–14 GHz band — from harmful interference, particularly under agenda item 1.2;
- The need to avoid interference from other services, particularly concerning satellite and VHF frequencies;
- IMO support for studies on the future development of VHF maritime radiocommunication, under agenda item 2.7 of WRC-31;
- A call for alignment with ITU Recommendations, maintaining focus on the operational integrity of existing maritime services.

DSC distress alerts by ships

NCSR 12 looked into draft modifications to MSC.1/Circ.1657 on *Procedures for responding to DSC distress alerts by ship*. The discussion focused on resolving discrepancies between the circular and Recommendation ITU-R M.541-11, which outlines operational procedures for digital selective calling equipment in the maritime mobile service. The revised circular also takes into account ITU-R Recommendation M.585-9, which specifies the priority and handling of maritime safety communications using DSC.

NCSR 12 agreed to the revised text for approval by MSC 111 and finalised a draft liaison statement to ITU-R, highlighting the need for consistency between IMO and ITU guidance. The Sub-Committee approved the liaison statement and requested the Secretariat to convey it to ITU, with endorsement of the action by MSC invited.

Search and Rescue (SAR)

Remotely Piloted Aircraft (RPA) and other autonomous craft

NCSR 12 confirmed that SAR services are not required to search for craft with no persons onboard. Distress beacons should not be used to alert for such craft.

Global SAR Plan

NCSR 12 developed a draft COMSAR circular on *Guidance for entering and updating information on search and rescue services into the Global SAR Plan and on how to get access to the information for operational use*, superseding COMSAR.1/Circ.55 and Circular Letter No.3588. Key elements include:

- Guidance for inputting and managing SAR data in GISIS, including rescue coordination centres (RCCs) and Telemedical Maritime Advice Services (TMAS);
- Provisions for entering geographical delimitations of SAR regions in line with GISIS constraints;
- Recommendations for ensuring periodic accuracy reviews; and
- Acknowledgement of the public utility of the Global SAR Plan by the wider maritime community.

NCSR 12 also recommended the development of a technical validation tool in the Global SAR Plan module in GISIS and invited Member States to continue contributing to improvements through the ICAO/IMO Joint Working Group.

Augmentation system in the World Wide Radio Navigation system (WWRNS)

An augmentation system is a system that provides correction and/or integrity information to improve the performance of existing GNSS signals. It does this by using data from ground reference stations, and may transmit enhancements via satellites or terrestrial means.

NCSR 12 agreed that augmentation systems are not part of the Worldwide Radionavigation System (WWRNS) for the purposes of IMO recognition, as they merely enhance existing GNSS attributes such as accuracy, integrity, availability, continuity, and reliability.

Instead of formal recognition, NCSR 12 developed a draft MSC resolution amending resolution A.1046(27) on *Worldwide Radionavigation Systems, introducing procedures and operational considerations for augmentation systems*. These systems may be used voluntarily by Administrations (flag State authority), without requiring inclusion in the IMO's list of recognized WWRNS components.

Key elements of the draft resolution are:

- Clarifies that augmentation systems are not standalone radionavigation systems, but support existing GNSS services;
- Provides a voluntary framework for Administrations to notify the IMO of the operational use of such systems;
- Emphasizes that use of augmentation services should not interfere with the core function of the primary GNSS systems;
- Stresses the need for transparency and continuity in service provision if these systems are referenced in navigation planning.

The scope of this work does not include performance standards for shipboard receivers, which will be addressed separately at NCSR 13.

VHF Data Exchange System (VDES)

VHF Data Exchange System (VDES) is a communication system intended to supplement AIS by providing a higher data capacity for maritime digital communications.

NCSR 11 (2024) agreed that the satellite segment of VDES was not yet sufficiently mature, and therefore focused only on SOLAS chapter V.

Liberia pointed out that, while not opposing the development of performance standards and guidance, amendments to SOLAS were not necessary at this stage due to the limited uptake of the technology and the administrative impact on survey checklists and documentation (e.g. replacing "AIS" with "AIS or VDES"). While there was several support to the issues raised by Liberia, NCSR 12 agreed to pursue amendments to SOLAS Chapter V based on the decision at NCSR 11.

NCSR 12 subsequently finalized the following for consideration by MSC 110 (scheduled for June 2025):

- Draft amendments to SOLAS chapter V and the HSC Code;
- A draft MSC resolution announcing potential implications for non-mandatory instruments, i.e. to read "AIS" with "AIS or VDES" without amending respective resolutions or circular;
- Draft performance standards for VDES shipborne equipment;
- Draft guidance for the installation of VDES.

Key leap from AIS by using VDES, while it is voluntary, are:

- Increased data throughput, allowing transmission of richer and more complex digital information;
- Expanded two-way communication, enabling more robust and flexible IP-based exchanges compared to AIS's limited short messaging capability;
- Dedicated spectrum channels for VDES, reducing congestion and enhancing prioritisation of safety-critical maritime data traffic.

Software maintenance

Having recalled the decisions of MSC 107 in relation to the inclusion of this output, NCSR 12 considered the Liberia co-sponsored document, proposing a draft MSC circular on guidelines for software maintenance of shipboard navigation and communication equipment and systems, aiming to improve the efficiency, effectiveness, safety and security of shipboard software maintenance by introducing a standardized, controlled and transparent process.

NCSR 12 developed a draft MSC circular for approval by MSC 111. Key points that are addressed in this circular are:

- Scope and terminology: The guidelines apply to shipboard computer-based equipment as defined under SOLAS chapters IV and V, with additional voluntary application to other systems listed in, such as dynamic positioning systems, power management systems, and surveillance systems;
- Software maintenance framework: Provisions cover planning, execution, certification, and documentation of software maintenance events. The use of certified service technicians and contracted service providers is clarified, and administrative burdens on crews are minimised;
- Logging and recordkeeping: A detailed onboard software log is required, along with structured electronic service reports, to ensure traceability and cybersecurity safeguards; and
- Clarifications: The draft circular confirms that it is non-mandatory and does not override existing provisions, including SOLAS regulation IV/15 (Maintenance of equipment) regarding shore-based maintenance.

Electronic Nautical Publication (ENP) (Agenda item 12)

NCSR 12 developed a draft MSC circular on *Guidelines for the carriage and use of electronic nautical publications (ENP) system*. Key points of the draft guidelines are:

- Provides unified implementation of SOLAS regulations V/19.2.1.4, V/19.2.1.5, and V/27 on the use of nautical publications onboard;
- Recognizes ENP as an acceptable electronic means of meeting carriage requirements for nautical publications, in lieu of hard copies;
- Clarifies that when nautical publication data is provided through Electronic Chart Display and Information System (ECDIS), these guidelines do not apply and do not override ECDIS performance standards;
- Requires appropriate backup arrangements when ENP is used to meet functional requirements.

Performance standards for the BeiDou satellite navigation system receiver

NCSR 12 revised the performance standards for shipborne BeiDou Satellite Navigation System (BDS) receiver equipment, set out in resolution MSC.379(93), for approval by MSC 111. The originally proposed change of geodetic datum from WGS 84 to the BeiDou Coordinate System (BDCS/CGCS2000) was withdrawn. Instead, WGS 84 was retained to ensure alignment with other positioning systems and ENC datums.

The revised resolution includes updated references to signal types, technical specifications, and installation criteria. It also harmonizes the application clause with IMO drafting guidance, specifying that the revised standards will apply to BDS equipment installed on or after 31 July 2028.

Data Distribution and Global IP-Based Connectivity for ECDIS S-100 Products

The advance feature of S-100 products is to present dynamic (live) updates of various nautical information through ENC/ECDIS. However, the methods and protocols for the connection remain undecided.

NCSR 12 held an initial discussion on the development of a draft MSC circular providing guidance on IP-based data exchange between shore-based facilities and ships. Key points addressed included:

- Clarification that SECOM (data exchange protocol designed by the International Hydrographic Organization (IHO)) and MCP (Maritime Connectivity Platform) are cited as examples only, preserving technology neutrality;
- Agreement that S-100 data, including navigational and meteorological warnings transmitted over IP-based services, do not constitute part of the GMDSS;
- Identification of the need for further work on the technical, operational, and procedural framework for secure data distribution.

As the text could not be finalised, NCSR agreed to continue work through a correspondence group coordinated by Australia, with interim results to be reported to the IMO/ITU Joint Experts Group in October 2025.

In addition, NCSR 12 encouraged all stakeholders, in particular IHO, IALA and IEC, to continue progressing the S-100-related work within their own domains to enable a timely establishment and operation of an S-100 framework for distribution of S-100 data to ships and to keep the IMO informed of their progress.

Any other business

Update on the progress of S-100 implementation

NCSR 12 considered the information provided on progress made with the development and implementation of the IHO Universal Hydrographic Data Model (S-100) intended for use in future S-100 ECDIS as base line standard for the next generation of Electronic Navigational Charts (S-101 ENCs) and other products and services in accordance with resolution MSC.530(106)/Rev.1. Following consideration, NCSR 12:

- .1 noted the commitment of IHO Member States to the implementation dates stated in resolution MSC.530(106)/Rev.1;
- .2 acknowledged the progress being made on IHO S-100 product specifications including the publication of the Operational Edition 2.0.0 of the Phase 1 Product Specifications; and
- .3 encouraged Member States to develop S-100 data set production and dissemination strategies.

Further information

For further information please contact: imo@lisr.com

Annex

Provisional list of draft resolutions and circulars

- Draft amendment on the Establishment of an Area to be Avoided (ATBA) off the coast of Réunion
- Draft amendment on the Revised mandatory ship reporting system in the Adriatic Sea (ADRIREP)
- Draft amendments to resolution MSC.314(88) on the Mandatory ship reporting system "SOUNDREP" in the Sound between Denmark and Sweden
- Draft amendments to resolution MSC.332(90) on the Mandatory ship reporting system "BELTREP" in the Storebælt (Great Belt) Traffic Area
- SN.1/Circ.272/Add.3, which updates water depth information and changes the geodetic datum used on the chart for the South Åland Sea deep-water route
- Draft MSC circular on establishment and operation of the IMO Terrestrial Broadcast Services Coordinating Panel
- Draft revision of resolution MSC.509(105) on dissemination of MSI and SAR-related information
- Draft MSC circular on Guidelines for the carriage and use of Electronic Nautical Publications (ENP) systems
- Draft revision of MSC.1/Circ.1657 on Procedures for responding to DSC distress alerts by ships
- Draft MSC circular on Guidelines for software maintenance of shipboard computer-based navigation and communication equipment and systems
- Draft MSC circular on Guidance for entering and updating information on search and rescue services into the Global SAR Plan
- Draft MSC resolution on recognition and use of augmentation systems in the Worldwide Radionavigation System (WWRNS)
- Draft amendments to SOLAS Chapter V and the HSC Code related to the implementation of VDES
- Draft MSC resolution addressing potential implications of VDES on non-mandatory instruments
- Draft performance standards for VDES shipborne equipment
- Draft guidance on the installation of VDES equipment
- Draft amendments to SOLAS regulations IV/5, V/4 and V/5 clarifying coastal State obligations to disseminate MSI through all recognized mobile satellite services (RMSSs)
- Draft revision to resolution MSC.379(93) on the performance standards for shipborne BeiDou Satellite Navigation System (BDS) receiver equipment